NTNU

Hybrid Analysis and Modeling for Digital Twins

Lecture 0 — what this course is, and why

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Outline

Digital twins: concept and capability

What stops the climb: challenges from industry

Three ways to build the model

The map: six parts

Three systems you will meet every week

How the course works

01

Digital twins: concept and capability

What a digital twin is

The term is used loosely enough to be a liability — to some a 3D visualisation, to others a data pipeline, to others a simulation model. One definition, used throughout this course:

Definition

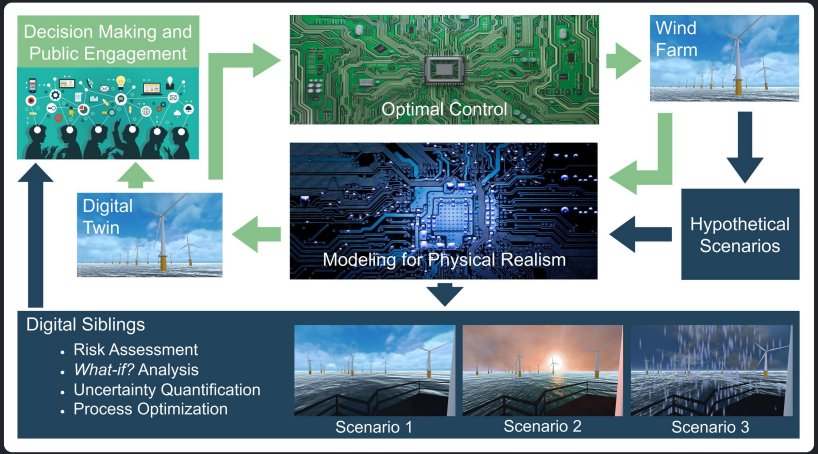
A **digital twin** is a virtual representation of a physical asset or process, enabled through data and simulators, for real-time prediction, optimisation, monitoring, control, and informed decision-making.

Real-time rules out anything that cannot keep pace with the asset — which is why we focus on making accurate models cheap in this course.

Informed rules out predictions with no indication of how much to trust them — which is why uncertainty is very important to state.

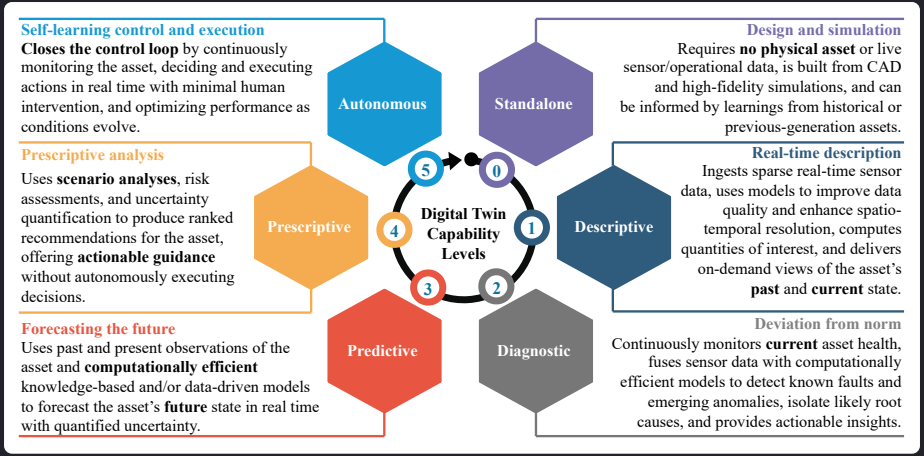
Two relatives: run the same models *offline* and you have a **digital sibling** (what-if, risk, UQ); archive their predictions across the asset's life and you have a **digital thread**, which informs the next design.

The digital twin as a working system



Two loops, *one* modelling block. Green: real time. Blue: offline what-if.

How capable is it? The 0–5 scale



Cumulative. Levels 0–2 describe the present; **3 is the first that says anything about the future**; 4–5 change what the asset does.

Level 0 — Standalone

Capability

- 5 Autonomous
- 4 Prescriptive
- 3 Predictive
- 2 Diagnostic
- 1 Descriptive
- **0 Standalone**

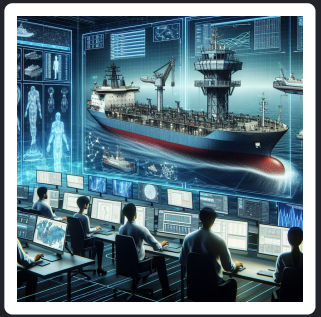


A description of the asset, disconnected from any real environment — the physical asset may not even exist yet.

Level 1 — Descriptive

Capability

- 5 Autonomous
- 4 Prescriptive
- 3 Predictive
- 2 Diagnostic
- **1 Descriptive**
- 0 Standalone



CAD models plus a real-time stream of sensor data describe the up-to-date state of the asset at any moment.

Level 2 — Diagnostic

Capability

- 5 Autonomous
- 4 Prescriptive
- 3 Predictive
- **2 Diagnostic**
- 1 Descriptive
- 0 Standalone



Presents diagnostic information — condition monitoring, fault isolation, troubleshooting.

Level 3 — Predictive

Capability

- 5 Autonomous
- 4 Prescriptive
- **3 Predictive**
- 2 Diagnostic
- 1 Descriptive
- 0 Standalone



Predicts future states or performance — the first rung that says anything about what has not happened yet.

Level 4 — Prescriptive

Capability

- 5 Autonomous
- **4 Prescriptive**
- 3 Predictive
- 2 Diagnostic
- 1 Descriptive
- 0 Standalone

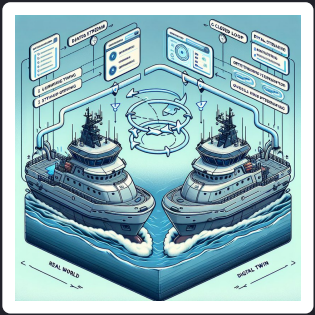


Recommends actions from what-if and risk analysis, each with quantified uncertainty. A human still decides.

Level 5 — Autonomous

Capability

- **5 Autonomous**
- 4 Prescriptive
- 3 Predictive
- 2 Diagnostic
- 1 Descriptive
- 0 Standalone



Closes the loop: the twin makes decisions and executes control actions on the asset autonomously.

Read the ladder again — it is a table of contents



The capability scale and the structure of this course are the *same list*:

- Levels 1–2 (describe, diagnose)
- Level 3 (predict)
- Levels 4–5 (prescribe, act)

Not a coincidence

The course is organised by what a twin *needs*, one rung at a time — not by what is mathematically tidy.

02

What stops the climb: challenges from industry

The evidence: an industry survey

Questionnaires went to operators, OEMs, software houses and classification bodies. **Twelve partners replied:**

Operators & energy

- ANEO
- Equinor
- Statkraft
- TotalEnergies
- SNSK

Maritime & subsea

- Kongsberg Maritime
- Vard
- 4Subsea
- Eidel

Software & data

- Kongsberg Digital
- Cognite
- Mainstream Renew. Power
- Norconsult

Class. & testing

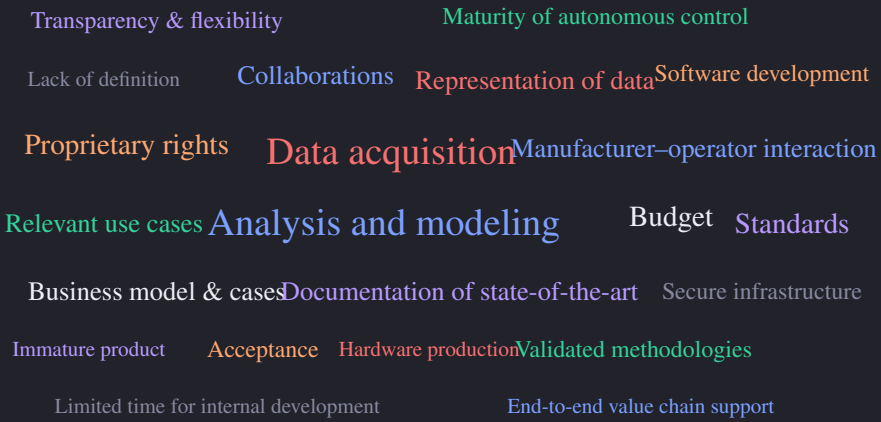
- DNV Maritime
- DNV Energy Sys.
- Force Technology

What the survey sought

A shared *terminology*, a *capability scale*, the *major challenges* — grouped as data, model and acceptance — and a map of *potential solutions*.

Stadtman et al. *Digital Twins in Wind Energy: Emerging Technologies and Industry-Informed Future Directions*. 2023

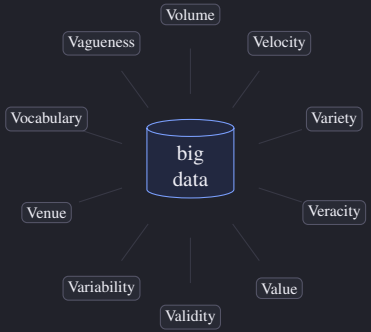
Challenges, in the industry's own words



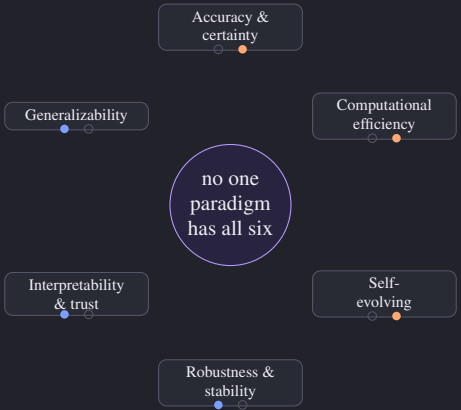
Data-related challenges

- Data **quality**, and no systematic collection
- **Sharing** across owners — silos, interoperability
- **Sparse** in space, uneven in time
- **Security**, and the choice of management stack
- A **talent** gap to run any of it

The famous “V”s are not a slogan — each is a distinct failure mode a twin must survive.



Model-related challenges — and why no single paradigm wins



A twin's model must be all six *at once*. But:

- Physics-based delivers generalisation, interpretability, stability.
- Data-driven delivers accuracy, efficiency, self-evolution.

Industrial acceptance — the last mile

Even a perfect model changes nothing if nobody adopts it:

- No consensus on what a twin *is*
- Thin awareness of the *value* it creates
- Technology readiness still *low*
- Missing *business models*
- Security, privacy, and IP projection



Four families of obstacle — and which are ours

Standards are institutional; data ownership is commercial. The model — and much of acceptance, through interpretability and honest uncertainty — is what a course can actually move.

Where academia can push hardest

Extract value from collected data

Integration with other systems

Applicable technologies, standards & platforms

Hybrid analysis and modeling

Wind-farm ontology

Optimal sensor placement

Forecasting

O&M applications

Data integration standards

Autonomy

Ease inspection & maintenance decisions

The single biggest recommendation to academia is, in as many words, **this course** — so that is where we go next.

03

Three ways to build the model

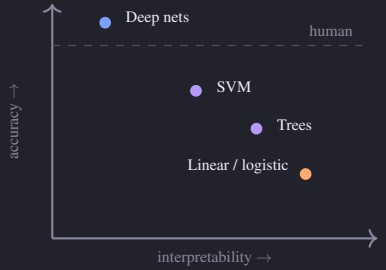
Two modeling cultures, honestly

	Physics-based (PBM)	**Data-driven (DDM)**
Comes from	first principles	measurements
Interpretable	yes — by construction	rarely
Extrapolates	as far as the assumptions	no (and silently)
Fast online	seldom	almost always
Needs data	little	lots
Fails by	missing physics, cost	distribution shift

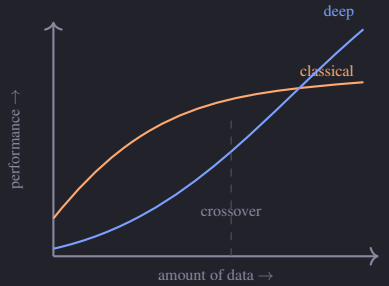
The thesis of this course

The two columns fail in *complementary* ways — so the interesting methods live between them, using physics for structure and data for what the physics gets wrong.

Why not just use deep learning?



More accurate is usually *less* interpretable.

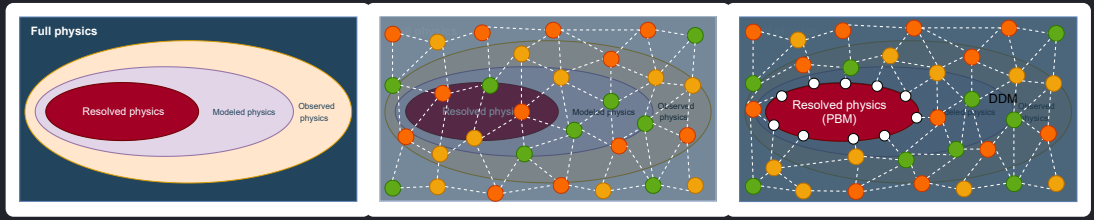


Deep learning overtakes — once data is plentiful.

The takeaway

Little data or a need to explain ⇒ classical methods abundant data ⇒ deep models. A twin needs both — and, where physics is known, something better than either.

The same argument, visually



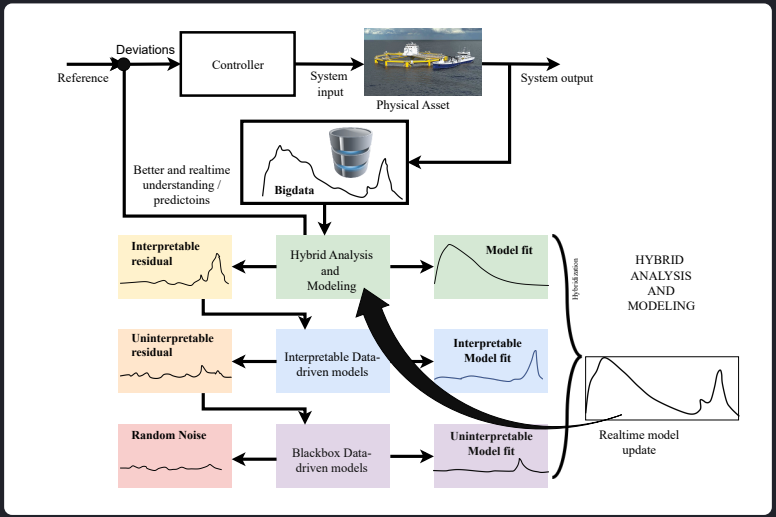
PBM resolves *part* of the physics

DDM covers what was *observed*

HAM keeps *both*

Resolved ⊂ modelled ⊂ observed ⊂ full physics. PBM owns the core and is mute outside it; DDM reaches everything measured but respects no structure; hybrids keep the core and let data cover the rest.

The big picture: Big Data Cybernetics



How hybridization works: squeeze the residual



Each stage removes the structure it can honestly explain and hands the leftover on. Physics first (most interpretable), a learned-but-interpretable model next, a black box last — and you *stop* when the residual is noise, because nothing interpretable is left to extract.

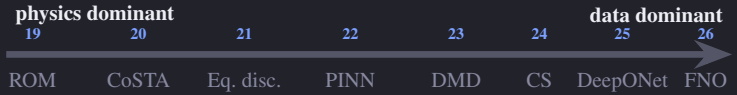
The two halves the title names

Hybrid *analysis*

The data-driven model dominates; domain knowledge makes its output mean something. The method finds the structure — the physics tells you which structure is a bearing fault and which is a change in sea state.

Hybrid *modelling*

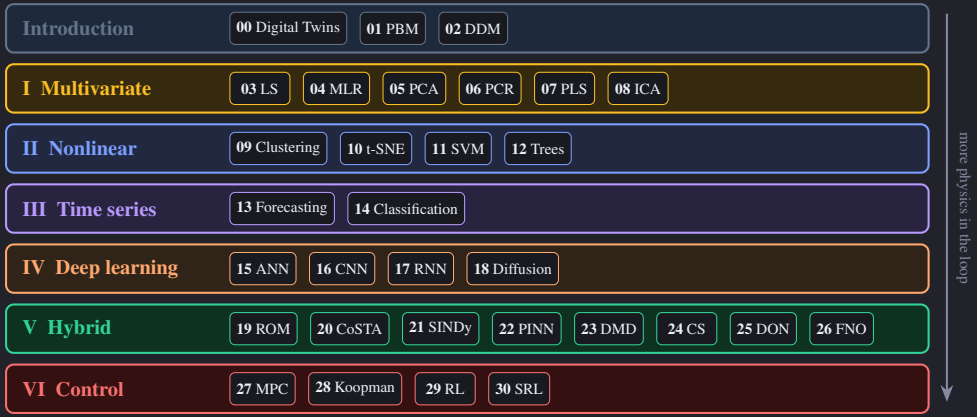
The physics-based model dominates; data supplies what the understanding lacks — a missing term, an unknown parameter, a closure, a basis, or the whole solution operator.



04

The map: six parts

The course map

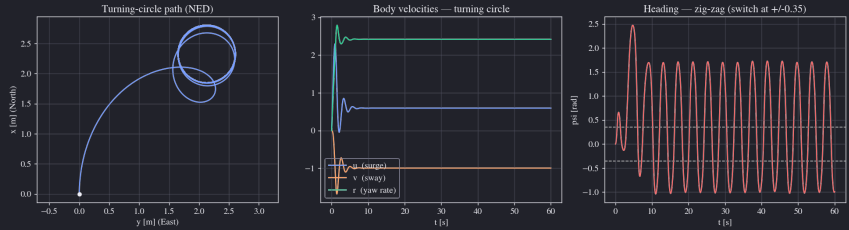


Six parts, one escalation: describe the data, then bend the description around nonlinearity, then take time seriously, then learn representations, then put the physics back in — and finally act on it.

05

Three systems you will meet every week

Running example 1: the autonomous surface vessel



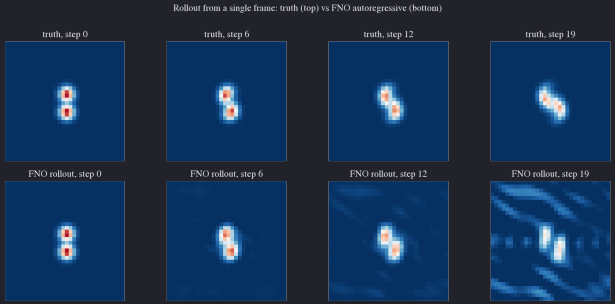
The baseline in action: a turning-circle *path* in the plane (left), the body velocities u, v, r settling to steady values (centre), and a *zig-zag* heading oscillation (right) — all from the one model below.

An **ODE** in time — the 3-DOF maneuvering model

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}.$$

- **Unknowns:** the hydrodynamic coefficients in $\mathbf{C}, \mathbf{D}$ are seldom known exactly — they must be identified from trials or data.
- **Efficiency:** control and what-if studies need the model to run far faster than real time.

Running example 2: heat and flow

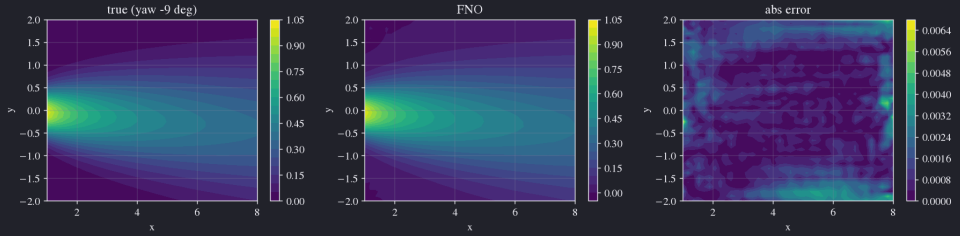


Two like-signed vortices rolled forward in time: the true flow (top) vs a neural-operator surrogate (bottom) — faithful for a dozen steps, then degrading.

PDEs in space and time: heat conduction (pure diffusion) and the Navier–Stokes equations (advection + diffusion, shown rolling a flow forward).

- **Efficiency:** resolving the fine scales makes a trustworthy solve slow — often far too slow for a twin.
- **Unknowns:** source terms, closures and boundary data are uncertain, and have to be corrected against measurements.

Running example 3: the wind turbine and its wake



A turbine *wake* — the velocity deficit trailing downstream: the true field (left), a neural-operator prediction (centre), and their small difference (right).

Both worlds at once: an **ODE** for the drivetrain and its control, and a **PDE** for the wake field (shown), fed by real operational data.

- **Unknowns:** the effective inertia, the blade-erosion drift and the wake-decay law are not known a priori — data has to supply them.
- **Efficiency:** a whole farm's wake fields are far too expensive to resolve directly for every layout or control decision.

06

How the course works

The course runs on `edaptic.no`

1. Go to **`www.edaptic.no`** and **create a login using your NTNU email**.
2. **Register for the course** `TTK29-2026`.
3. All the **lecture notes and materials** live there — and the **assignments** are posted there too.

Good practice

Complete each **assignment before moving on** to the next topic — the assignments are where the ideas become yours.

Feedback

Leave your feedback on the course and the platform **directly on edaptic**, whenever something helps or gets in the way.

Anatomy of the materials

Artifact	Use it for	Variants
Slides	the lecture	clean / with notes on edaptic
Notebooks	every figure, every number	Try out to understand the lecture
Assignments	automatically graded	elevate if not satisfied
Projects	open-ended practice	Choose the problem as you like

The reproducibility contract

Every figure on every slide is generated by a notebook in the repository. If you can run `assignment_setup.ipynb`, you can rebuild the entire course from source.

Getting set up (this week)

1. Install the environment: `conda env create -f python/environment.yml`
2. Open `python/00-introduction/notebooks/` and run `intro_three_paradigms.ipynb` — the figure from today.
3. Submit **Assignment 0** (`assignment_setup.ipynb`): four asserts that verify your stack. Worth 0 points and all of your week-3 sanity.

Notation, fixed for the whole course

scalar x (italic) vector $\boldsymbol{x}$ (bold lowercase; $\boldsymbol{\theta}$ for Greek) matrix $\mathbf{A}$ (bold capital) — written out in full, never hidden behind macros.